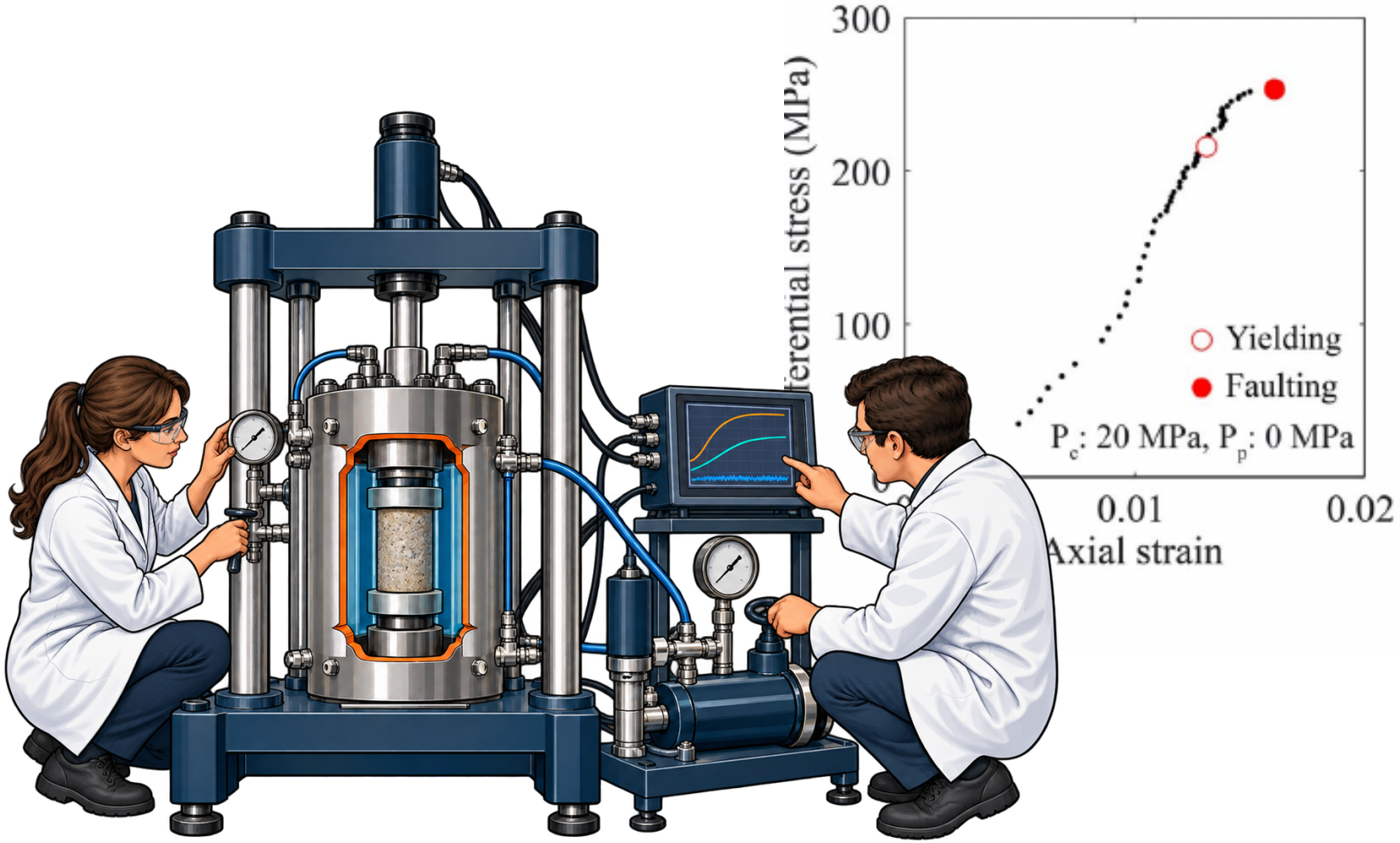
A loading test produces interpretable data



One test, many quantities

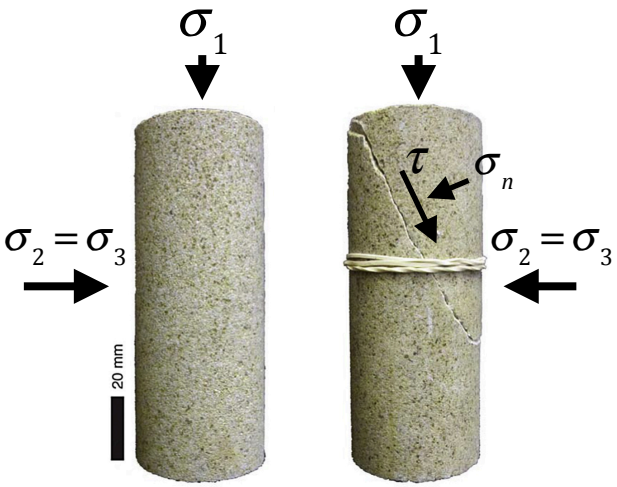
Geometry, loading, shortening and wave travel times become data we can compare.

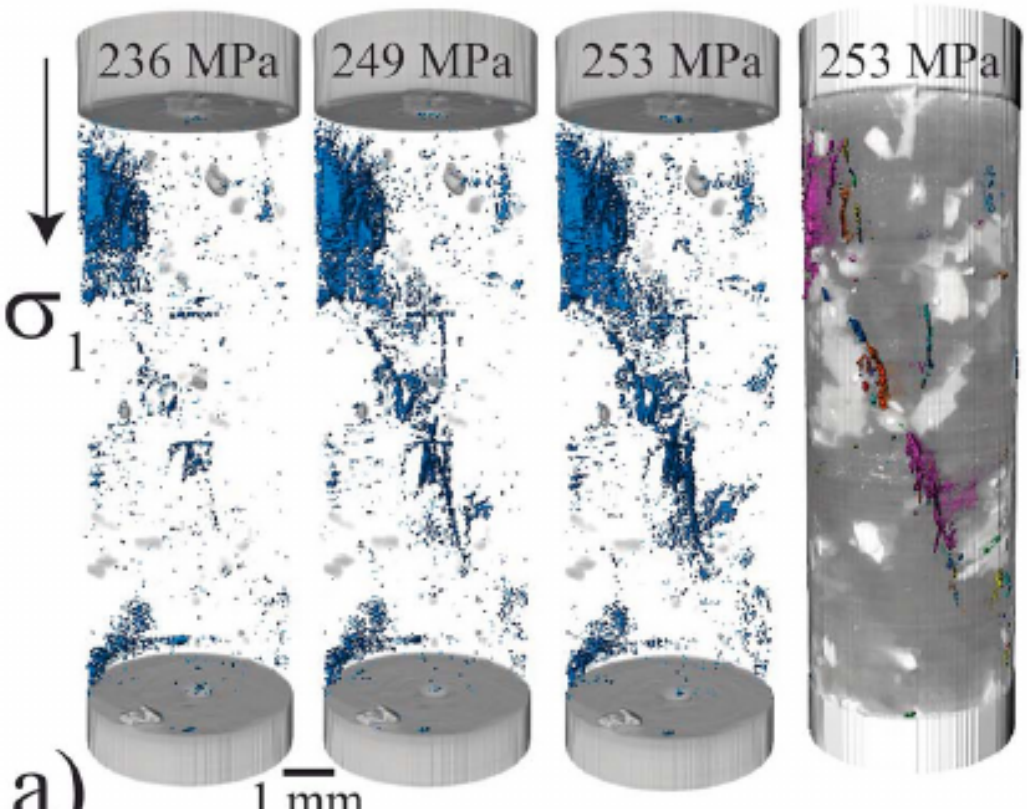
Laboratory 6 → analysis 7

One experiment yields several measurements

MECHANICAL	Force and displacement become stress and strain.
ULTRASONIC	Arrival time and path length become wave speed.
METADATA	Sample size, settings and calibration make results traceable.

Damage develops before the final fracture





What can we detect?

Curvature in the stress–strain record

Changes in ultrasonic waves

Acoustic emission activity

Measure damage, then defend an interpretation

LECTURE 12

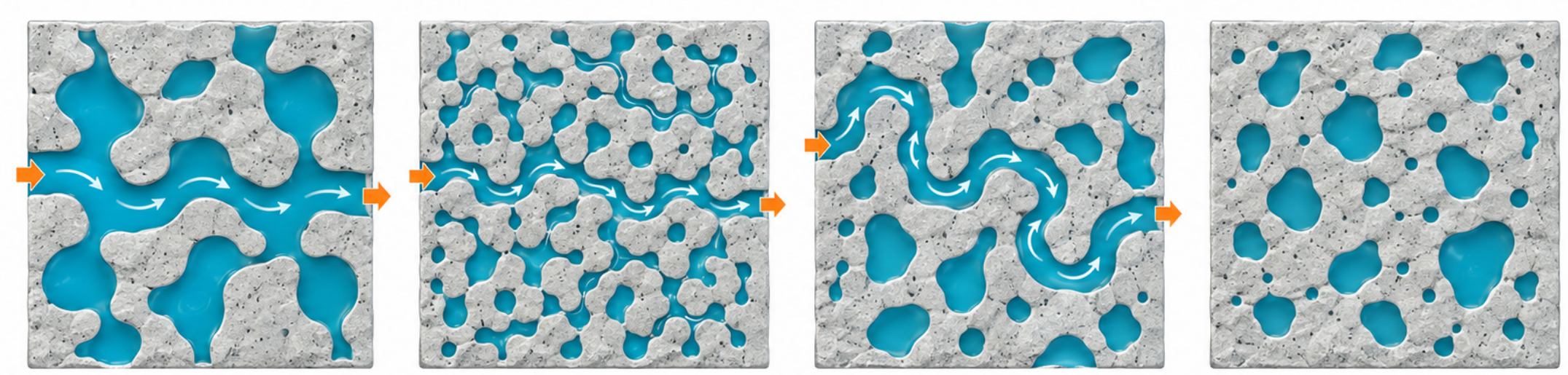
Collect mechanical and acoustic emission records.

LECTURE 13

Reduce, align and plot the records. Decide which changes identify stages of damage.

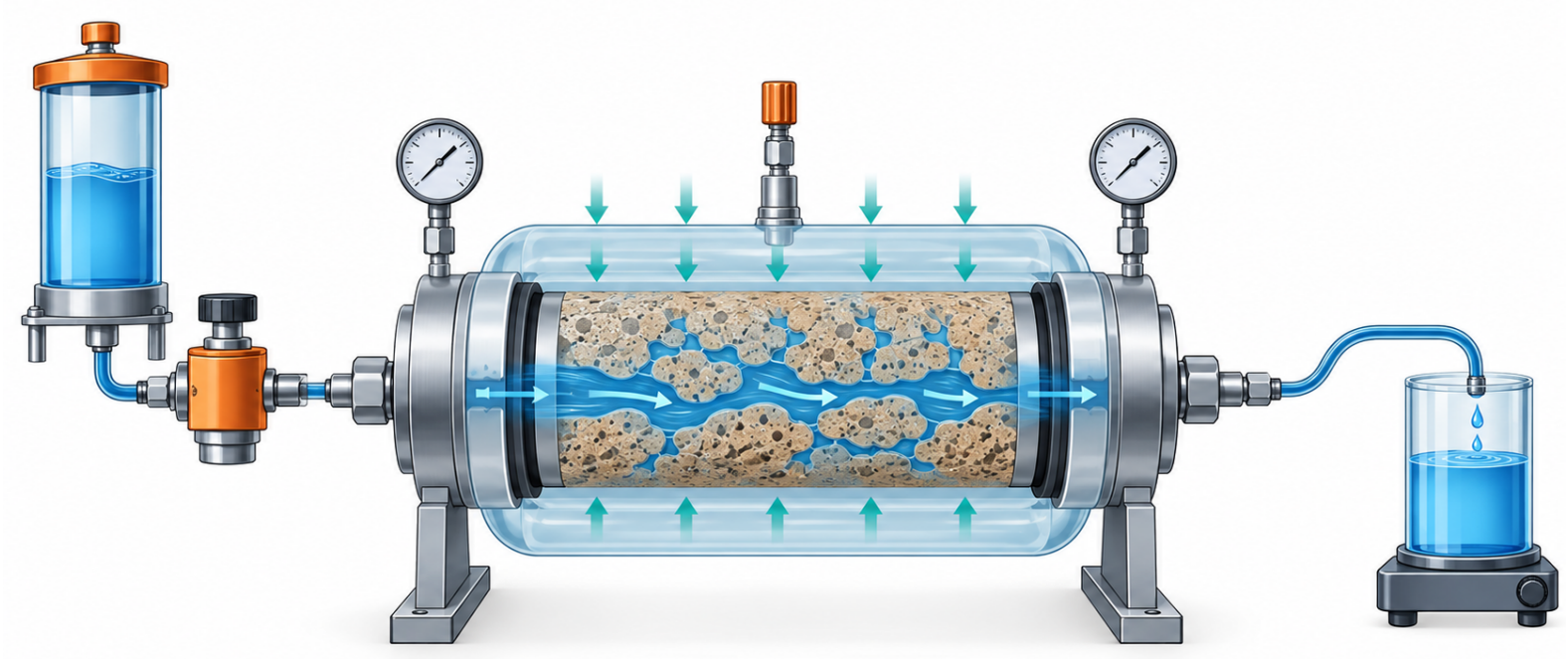
A plot is evidence only when we can explain how it was produced.

Equal porosity can hide very different flow



Which pore network would transmit water and pressure most readily?

A pressure difference drives flow through a sample



What must we know to infer permeability from measured flow?

Which links could the course help you test?

I Characterize the rock and its mechanical response

II Recognize damage and assess failure conditions

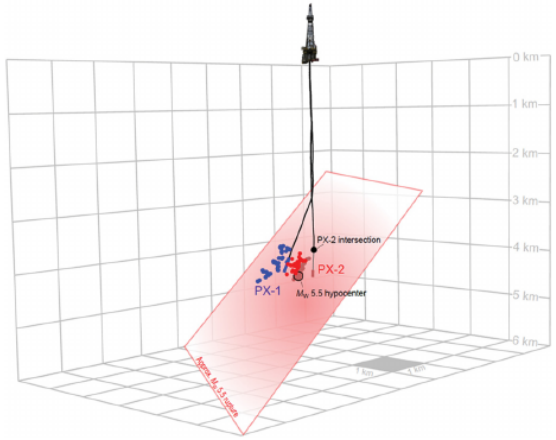
III Estimate how fluids and pressure could affect the system

The field case also needs fault geometry and seismic observations.

Mw 5.5

An earthquake occurred near an enhanced geothermal project after hydraulic stimulation.

What evidence could link the operations to the earthquake?



Prediction, measurement and interpretation

01	**PREDICT**	What should happen?
02	**OBSERVE**	What did we record?
03	**EXPLAIN**	Which model fits?
04	**REVISE**	What remains uncertain?